

Applied Class 10 – Solutions

Part 1

Lung function data for 25 cystic fibrosis patients in a study by O'Neill *et al.* (1983) was presented in Douglas Altman's *Practical Statistics for Medical Research* (1991). The data contains variables recorded for each patient and here we are interested in the relationship between two of these, **weight** (weight, kg) and **pemax** (maximum expiratory pressure, cm of H₂O).

- a) Write down the linear model and its assumptions to study the relationship between the response variable, **pemax**, and the explanatory variable, **weight**.

$Y \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma)$ where $Y = \text{pemax}$ and $X = \text{weight}$.

That is, we assume that the mean **pemax** comes from a *linear relationship* with **weight**, with variability about the line that is *Normal* and with *constant standard deviation* (σ).

- b) The R output for the linear model is presented below. Compute the values of A and C , and then give a value for B .

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	63.5456		A	5.003 4.63e-05 ***
weight	1.1867	0.3009	3.944	B

Residual standard error: 26.38 on C degrees of freedom

$$C = n - 2 = 25 - 2 = 23.$$

$$A = \text{se}(b_0). \text{ Since } t = (b_0 - 0)/\text{se}(b_0), \text{ we know } 5.003 = (63.5456 - 0)/A, \text{ so } A = \frac{63.5456}{5.003} = 12.70.$$

$$B = 2P(t_{23} \geq 3.944). \text{ Using R, } B = 2 \times 0.000323 = 0.000646.$$

- c) Based on the linear model, what is the estimated mean maximum expiratory pressure for patients with a weight of 50 kg?

Regression equation is $\hat{y} = 63.5456 + 1.1867x$. Substituting $x = 50$ gives

$$\hat{y} = 63.5456 + 1.1867 \times 50 = 122.88 \text{ cm H}_2\text{O}$$

- d) Conduct a hypothesis test to determine whether there is any evidence of a linear association between **pemax** and **weight**. What do you conclude?

Test $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$. From the table in (b), the p -value is 0.000646.

Thus, we have strong evidence to suggest that there is a linear relationship between **pemax** and **weight**.

Part 2

A study reported on the survival rate among patients suffering cardiac arrest both when resuscitation was started by (trained) lay people and when it was delayed until the arrival of an ambulance crew. A total of 27 patients survived out of the 75 attended by lay people, while 130 survived out of the 556 attended by an ambulance crew.

- a) Give a 95% confidence interval for the difference between p_1 and p_2 , where p_1 denotes the proportion of successful resuscitations in the lay-trained class and p_2 the proportion of delayed successful resuscitations by ambulance crews.

$$\hat{p}_1 = \frac{27}{75} = 0.36 \qquad \hat{p}_2 = \frac{130}{556} = 0.2338$$

A 95% CI for $(p_1 - p_2)$ is given by $(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

$$\begin{aligned} \text{Here this is } & (0.36 - 0.2338) \pm 1.96 \sqrt{\frac{0.36(1-0.36)}{75} + \frac{0.2338(1-0.2338)}{556}} \\ & = 0.1262 \pm 1.96 * 0.05826 \\ & = 0.1262 \pm 0.114189, \text{ or } (0.0120, 0.2404). \end{aligned}$$

Thus, we are 95% confident that the proportion of successful resuscitations is between 1.2% and 24% higher in the lay-trained class.

- b) Based on the confidence interval, is there any evidence of a difference in the proportion of successful resuscitations in the lay-trained class compared to the proportion of delayed successful resuscitations by ambulance crews?

95% CI for $(p_1 - p_2)$ does not include 0 so we have evidence to conclude that there is a difference in the two population proportions.

- c) How would the analysis need to change if the researchers were aiming to show that a program to train lay people to perform resuscitation would be beneficial?

We need to perform a one-sided hypothesis test, as follows:

$$H_0: p_1 = p_2 \text{ vs } H_1: p_1 > p_2$$

d) What is the p -value for the hypothesis test in part (c)? What do you conclude?

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}} = \frac{(0.36 - 0.2338) - 0}{0.05826} = 2.166152$$

The p -value = 0.015, giving moderate evidence to conclude that a program to train lay people to perform resuscitation would be beneficial.

You may find the following output from R useful in answering the questions on this sheet.

```
> 2*pt(-3.944,df=25)
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```
[1] 0.0005720022
```

```
> 1-pt(3.944,df=23)
```

```
[1] 0.000323207
```

```
> 2*pt(-3.944,df=23)
```

```
[1] 0.0006464141
```

```
> pnorm(2.166)
```

```
[1] 0.9848444
```

```
> pnorm(-1.96)
```

```
[1] 0.0249979
```